

The Convex Hull Of A Set Of Vectors

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Journal of Undergraduate Mathematics
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Preface

The convex hull of a set of vectors has many interesting properties in Euclidean and unitary spaces. In this monograph, we demonstrate the use of vector algebra in studying higher dimensional analogues of familiar properties of triangles in the plane. The special beauty of the study of plane figures comes from the ability to draw accurate diagrams and use natural geometric insight. In higher dimensions, the geometric insight is not as well-developed and often the natural generalizations need modification.

The table of contents gives a list of properties considered. Each topic is presented in the plane; in finite dimensional Euclidean spaces; in unitary spaces; and in spaces of linear transformations. Many of Euclid's ideas are used to prove results concerning triangles. This monograph contains 49 exercises, many of which have several parts. They range in difficulty from routine to original research problems. In most exercises the intention is to investigate the statement, i.e., find counterexamples; find conditions under which it holds; etc. Some exercises may be false as stated. We are hopeful that the reader will be able to find many research problems which we have not listed in the exercises.

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1 Preliminaries

A vector space over the field of real numbers for which an inner product is given is usually called a Euclidean space. We are interested in finite dimensional vector spaces and will use R_n to denote the usual Euclidean space of dimension n . Vectors and scalars will be denoted by small Greek and Latin letters, respectively. The inner product of ξ and η will be denoted by (ξ, η) , and the norm of ξ is defined by $\|\xi\| = (\xi, \xi)^{1/2}$.

We will assume that the reader is familiar with basic topics of linear algebra such as linear independence, span, basis, orthogonality, etc. We will also assume familiarity with standard theorems of linear algebra and matrix theory.

For a set of vectors $(\xi_1, \dots, \xi_k)$ in R_n , we let $[\xi_1, \dots, \xi_k]$ denote the subspace of R_n spanned by this set. A *k-dimensional plane* in R_n is a set of vectors of the form

$$\xi_0 + [\xi_1, \dots, \xi_k]$$

where $\xi_1, \dots, \xi_k \in R_n$ and $\{\xi_1, \dots, \xi_k\}$ is a linearly independent set in R_n . For a set of vectors $\{\eta_0, \dots, \eta_m\}$ in R_n , let $P(\eta_0, \dots, \eta_m)$ denote the plane determined by $\eta_0, \dots, \eta_m$, i.e.,

$$P(\eta_0, \dots, \eta_m) = \eta_0 + [\eta_1 - \eta_0, \dots, \eta_m - \eta_0]$$

The dimension of $P(\eta_0, \dots, \eta_m)$ is the dimension of subspace $[\eta_1 - \eta_0, \dots, \eta_m - \eta_0]$ of R_n .

For a set of vectors $\{\eta_1, \dots, \eta_m\}$ in R_n , let $H(\eta_1, \dots, \eta_m)$ denote the convex hull of $\{\eta_1, \dots, \eta_m\}$, i.e., $H(\eta_1, \dots, \eta_m)$ is the set of all vectors $\xi \in R_n$ such that

$$\xi = \sum_{i=1}^m t_i \eta_i \text{ with } t_i \geq 0 \text{ and } \sum_{i=1}^m t_i = 1$$

The *Gram matrix* of the ordered set of vectors $\{\xi_1, \dots, \xi_k\}$ in R_n is the $k \times k$ matrix

$$G = \begin{pmatrix} (\xi_1, \xi_1) & \cdots & (\xi_1, \xi_k) \\ \vdots & \ddots & \vdots \\ (\xi_k, \xi_1) & \cdots & (\xi_k, \xi_k) \end{pmatrix}$$

1.1 Theorem

If $\{\xi_1, \dots, \xi_k\}$ is a linearly independent set in R_n , then the Gram matrix of $\{\xi_1, \dots, \xi_k\}$ is nonsingular.

Proof: Let $\{\alpha_1, \dots, \alpha_k\}$ be an orthonormal basis of $[\xi_1, \dots, \xi_k]$, and let a_{im} for $i, m = 1, \dots, k$ be the real numbers uniquely defined by

$$\xi_i = \sum_{m=1}^k a_{im} \alpha_m \text{ for } i = 1, \dots, k$$

For any $i, j = 1, \dots, k$, we have

$$(\xi_i, \xi_j) = \left(\sum_m a_{im} \alpha_m, \sum_\ell a_{j\ell} \alpha_\ell \right) = \sum_{m=1}^k a_{im} a_{jm}$$

and hence $G = AA'$, where A is the $k \times k$ matrix

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & \ddots & \vdots \\ a_{k1} & \cdots & a_{kk} \end{pmatrix}$$

and A' denotes the transpose (or adjoint) of A . Since A has a linearly independent set of rows, it is nonsingular, and thus G is also nonsingular.

1.2 Corollary

The Gram matrix of a set of linearly independent vectors is positive definite.

Proof: Consider $G = AA'$ as in the proof of Theorem 1.1. The matrix G is self-adjoint (or hermetian) as

$$G' = (AA')' = AA' = G$$

For each $\xi \in R_k$ we have

$$(G\xi, \xi) = (AA'\xi, \xi) = (A'\xi, A'\xi) = \|A'\xi\|^2$$

Since A' is nonsingular, we conclude that $(G\xi, \xi) > 0$ if $\xi \neq 0$ (the zero vector). Hence G is positive definite.

Exercises

1. Prove the converse of Theorem 1.1.
2. Show that the Gram matrix of *any* set of vectors $\{\xi_1, \dots, \xi_k\}$ in R_n is non-negative.
3. Show that the determinant of the Gram matrix of $\{\xi_1, \dots, \xi_k\}$ is independent of the ordering of $\xi_1, \dots, \xi_k$.
4. Let $\{\xi_1, \dots, \xi_k\}$ be a linearly independent set in R_n and let $\{\alpha_1, \dots, \alpha_k\}$ be the orthogonal set obtained from $\{\xi_1, \dots, \xi_k\}$ by the Gram-Schmidt orthogonalization process. Show that the determinants of the Gram matrices of $\{\xi_1, \dots, \xi_k\}$ and $\{\alpha_1, \dots, \alpha_k\}$ are equal.

2 Volumes

The Gram matrix plays an important role in volume measures. The k -dimensional volume measure of a subset of R_n is defined analytically (see Smith [5, p.369]) and has the following property. The k -dimensional volume measure of the parallelepiped defined by the linearly independent set $\{\xi_1, \dots, \xi_k\}$ in R_n is found by expressing $\xi_1, \dots, \xi_k$ in terms of an orthonormal basis $\{\alpha_1, \dots, \alpha_k\}$ of $[\xi_1, \dots, \xi_k]$ and then taking the absolute value of the determinant of the resulting coefficient matrix. In the notation of Section 1, we see that the k -dimensional volume measure of the parallelepiped defined by $\{\xi_1, \dots, \xi_k\}$ is $V = |\det A|$. We also see from the proof of Theorem 1.1 that $\det G = (\det A)^2$, and hence conclude that

$$V = (\det G)^{\frac{1}{2}}$$

(The determinant of G is called the *Gram determinant* of $\{\xi_1, \dots, \xi_k\}$.)

We will examine the case $n = k = 2$ in more detail. Let $\{\xi_1, \xi_2\}$ be a linearly independent set in R_2 . The parallelogram defined by this set is $H(0, \xi_1, \xi_2, \xi_1 + \xi_2)$ (Figure 2.1).

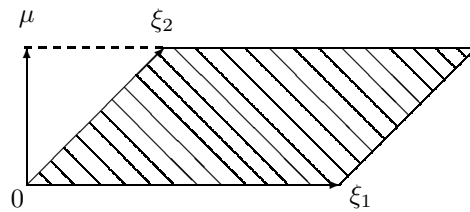


Figure 2.1

If η is the projection of ξ_2 into the orthogonal complement of $[\xi_1]$, then the area of this parallelogram is

$$A = \|\xi_1\| \|\eta\| = \|\xi_1\| \|\xi_2\| \sin t$$

where t is the angle between ξ_1 and ξ_2 . Thus

$$A^2 = \|\xi_1\|^2 \|\xi_2\|^2 (1 - \cos^2 t) = \|\xi_1\|^2 \|\xi_2\|^2 - (\xi_1, \xi_2) = \det \begin{pmatrix} (\xi_1, \xi_1) & (\xi_1, \xi_2) \\ (\xi_2, \xi_1) & (\xi_2, \xi_2) \end{pmatrix}$$

To generalize the ideas just considered, let $\{\xi_1, \dots, \xi_k\}$ be a set of linearly independent vectors in R_n . Fix an integer m with $0 < m < k$, let $M = [\xi_1, \dots, \xi_m]$, and let P denote the orthogonal projection of R_n onto $M^\perp$ (the orthogonal complement of M in R_n). We claim that $\{P\xi_{m+1}, \dots, P\xi_k\}$ is a linearly independent set in R_n . Indeed, if $a_{m+1}, \dots, a_k$ are real numbers such that

$$a_{m+1}P\xi_{m+1} + \dots + a_kP\xi_k = 0$$

then $P(a_{m+1}\xi_{m+1} + \dots + a_k\xi_k) = 0$ and hence $a_{m+1}\xi_{m+1} + \dots + a_k\xi_k \in M$. It follows from the linear independence of $\{\xi_1, \dots, \xi_m\}$ and the definition of M , that $a_{m+1}\xi_{m+1} + \dots + a_k\xi_k = 0$, and hence that $a_{m+1} = \dots = a_k = 0$.

We therefore obtain two new parallelepipeds - the parallelepiped defined by $\{\xi_1, \dots, \xi_m\}$, which is called the *base parallelepiped*, and the parallelepiped defined by $\{P\xi_{m+1}, \dots, P\xi_k\}$, which is called the *altitude parallelepiped*.

2.2 Theorem

In the preceding setting, the k -dimensional volume measure of the parallelepiped defined by $\{\xi_1, \dots, \xi_k\}$ is $V = V_1 V_2$, where V_1 is the m -dimensional volume measure of the base parallelepiped and V_2 is the $(k - m)$ -dimensional volume measure of the altitude parallelepiped.

Proof: Since $[\xi_1, \dots, \xi_m] \cap [P\xi_{m+1}, \dots, P\xi_k] = 0$, it follows from the linear independence of $\{\xi_1, \dots, \xi_m\}$ and $\{P\xi_{m+1}, \dots, P\xi_k\}$ that $\{\xi_1, \dots, \xi_m, P\xi_{m+1}, \dots, P\xi_k\}$ is a linearly independent set in R_n . Let $\{\alpha_1, \dots, \alpha_k\}$ be an orthonormal basis of $[\xi_1, \dots, \xi_k]$ such that $\alpha_1, \dots, \alpha_m \in M$. Writing $\xi_1, \dots, \xi_k$ in terms of $\alpha_1, \dots, \alpha_k$, we see that $V = |\det B|$, where

$$B = \begin{pmatrix} \begin{bmatrix} (\xi_1, \alpha_1) & \dots & (\xi_1, \alpha_m) \\ \vdots & & \vdots \\ (\xi_m, \alpha_1) & \dots & (\xi_m, \alpha_m) \end{bmatrix} & \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \\ \begin{bmatrix} (\xi_{m+1}, \alpha_1) & \dots & (\xi_{m+1}, \alpha_m) \\ \vdots & & \vdots \\ (\xi_k, \alpha_1) & \dots & (\xi_k, \alpha_m) \end{bmatrix} & \begin{bmatrix} (\xi_{m+1}, \alpha_{m+1}) & \dots & (\xi_{m+1}, \alpha_k) \\ \vdots & & \vdots \\ (\xi_k, \alpha_{m+1}) & \dots & (\xi_k, \alpha_k) \end{bmatrix} \end{pmatrix}$$

Letting

$$C = \begin{pmatrix} (\xi_1, \alpha_1) & \cdots & (\xi_1, \alpha_m) \\ \vdots & & \vdots \\ (\xi_m, \alpha_1) & \cdots & (\xi_m, \alpha_m) \end{pmatrix}$$

and

$$D = \begin{pmatrix} (\xi_{m+1}, \alpha_{m+1}) & \cdots & (\xi_{m+1}, \alpha_k) \\ \vdots & & \vdots \\ (\xi_k, \alpha_{m+1}) & \cdots & (\xi_k, \alpha_k) \end{pmatrix}$$

we see that $\det B = \det C \det D$ and $V_1 = |\det C|$. Since

$$P\xi_j = (\xi_j, \alpha_{m+1})a_{m+1} + \cdots + (\xi_j, \alpha_k)a_k \text{ for } j > m$$

we have $V_2 = |\det D|$ and hence $V = V_1 V_2$ as desired.

Note that Birkhoff and MacLane [3, p.308] *define* the k -dimensional volume measure of a parallelepiped inductively so that Theorem 2.2 holds. That is, they define the k -dimensional volume measure of the parallelepiped defined by $\{\xi_1, \dots, \xi_k\}$ to be $V = V_1 V_2$ where V_1 is the $(k-1)$ -dimensional volume measure of the base parallelepiped and V_2 is the one dimensional volume measure (or length) of the altitude parallelepiped. Using this definition, Birkhoff and MacLane show that V is given by (2.1).

We begin the study of volumes of convex hulls by considering the area of a triangle. The triangle defined by the linearly independent set $\{\xi_1, \xi_2\}$ in R_2 is $H(o, \xi_1, \xi_2)$ (Figure 2.2). Let $\xi_1 = (x_1, y_1)$ and $\xi_2 = (x_2, y_2)$ with respect to the usual coordinates on R_2 .

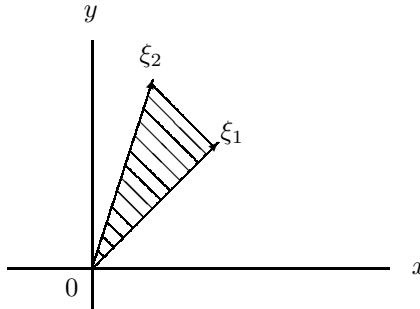


Figure 2.2

If we employ a double integral for computing the area of this triangle, we must describe the domain of integration D in two parts. However, if we use the transformation

$$\begin{cases} x = x_1 t_1 + x_2 t_2 \\ y = y_1 t_1 + y_2 t_2 \end{cases}$$

in the change of variable theorem, we obtain the area as

$$A = \int \int_D dx dy = |\det A| \int_0^1 \int_0^{1-t_1} dt_2 dt_1 = \frac{|\det A|}{2}$$

where $A = \begin{pmatrix} x_1 & x_2 \\ y_1 & y_2 \end{pmatrix}$.

We can now generalize this result to n dimensions. The reader should verify by induction that

$$\int_0^1 \int_0^{1-t_1} \cdots \int_0^{1-t_1-\cdots-t_{n-1}} dt_n \cdots dt_2 dt_1 = \frac{1}{n!}$$

Suppose now that $\{\xi_1, \dots, \xi_n\}$ is a linearly independent set in R_n , and let $\xi \in H(0, \xi_1, \dots, \xi_n)$, i.e.,

$$\xi = \sum_{i=1}^n t_i \xi_i \text{ with } t_i \geq 0 \text{ and } \sum_{i=1}^n t_i \leq 1$$

Let $\xi = (x_1, \dots, x_n)$ and $\xi_i = (x_{i1}, \dots, x_{in})$ for $i = 1, \dots, n$ with respect to the usual coordinates on R_n . We now have

$$x_i = \sum_{j=1}^n x_{ij} t_j \text{ for } i = 1, \dots, n$$

which we consider as a transformation in the change of variable theorem.

The n -dimensional volume measure of $D = H(0, \xi_1, \dots, \xi_n)$ is

$$V = \int_D \cdots \int dx_1 \cdots dx_n = |\det A| \int_0^1 \cdots \int_0^{1-t_1-\cdots-t_{n-1}} dt_n \cdots dt_1$$

where

$$A = \begin{pmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{nn} \end{pmatrix}$$

Once again we use the fact that the Gram determinant of $\{\xi_1, \dots, \xi_n\}$ is equal to $(\det A)^2$ and hence conclude that the n -dimensional measure of $H(0, \xi_1, \dots, \xi_n)$ is

2.3 Equation

$$V = \frac{(\det G)^{\frac{1}{2}}}{n!}$$

More generally, one can show (see Smith [5, p.371]) that if $\xi_1, \dots, \xi_k \in R_n$, then the k -dimensional volume measure of $H(0, \xi_1, \dots, \xi_k)$ is the square root of the Gram determinant of $\{\xi_1, \dots, \xi_k\}$ divided by $k!$.

Exercises

1. Define the parallelepiped defined by $\{\xi_1, \dots, \xi_k\}$ in R_n as the convex hull of a certain set of vectors.

2. Generalize the statement of Theorem 2.2 to the case where the parallelepiped defined by $\{\xi_1, \dots, \xi_k\}$ is split into r parallelepipeds with $2 \leq r \leq k$. Note that when $r = k$, one obtains the volume as a product of lengths.
3. Let M be a subspace of R_n and let P denote the orthogonal projection onto M . For any set of vectors $\{\xi_1, \dots, \xi_k\}$ in R_n , prove that the Gram determinant of $\{P\xi_1, \dots, P\xi_k\}$ is less than or equal to the Gram determinant of $\{\xi_1, \dots, \xi_k\}$. Give a geometric interpretation of this fact.
4. Let $\{\xi_1, \xi_2\}$ be a linearly independent set of vectors in R_3 , let M_1, M_2, M_3 be three mutually orthogonal 2-dimensional subspaces of R_3 , and let P_1, P_2, P_3 be the orthogonal projections of R_3 onto M_1, M_2, M_3 , respectively. Let $A, A_i (i = 1, 2, 3)$ denote the areas of the parallelograms defined by the sets $\{\xi_1, \xi_2\}, \{P_i\xi_1, P_i\xi_2\} (i = 1, 2, 3)$, respectively. Prove that
 - (a) $A_1^2 + A_2^2 \leq A^2$ (*Bessel's inequality*),
 - (b) $A_1^2 + A_2^2 + A_3^2 = A^2$ (*Pythagorean identity*)

Investigate some of the many generalizations to R_n .

3 Perpendicular Equisectors

If $\alpha, \beta \in R_2$ and $\alpha \neq \beta$, then a vector ξ is a *perpendicular bisector* of the line segment which connects α and β if $(\xi - \mu, \alpha - \beta) = 0$, where $\mu = \frac{(\alpha + \beta)}{2}$. Note that the set of perpendicular bisectors of the line segment which connects α and β is the line through the midpoint μ of that segment which is perpendicular to $\alpha - \beta$.

3.1 Theorem

A vector $\xi \in R_2$ is a perpendicular bisector of the line segment determined by α and β if and only if $\|\xi - \alpha\| = \|\xi - \beta\|$ (Figure 3.1).

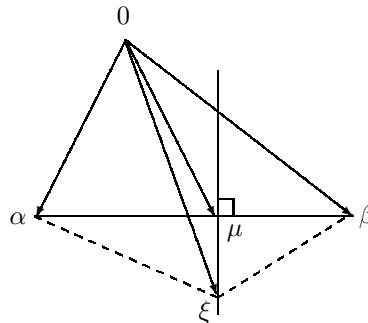


Figure 3.1

Proof: The result follows from the relationship

$$2(\xi - \mu, \alpha - \beta) = 2(\xi, \alpha) - 2(\xi, \beta) - \|\alpha\|^2 + \|\beta\|^2 = \|\xi - \alpha\|^2 - \|\xi - \beta\|^2$$

3.2 Theorem

There is a unique perpendicular bisector of the three sides of a triangle.

Proof: Let $\{\alpha, \beta\}$ be a linearly independent set in R_2 and consider the triangle $H(0, \alpha, \beta)$. Since $\{\alpha, \beta\}$ is a linearly independent set, the lines of perpendicular bisectors of the segments determined by 0 and α and by 0 and β are not parallel. In fact, the angle between these lines is the smaller of the angles between α and β and between α and $-\beta$. It now follows from Theorem 3.1 that there is a unique vector $\xi \in R_2$ such that

$$\|\xi\| = \|\xi - \alpha\| \text{ and } \|\xi\| = \|\xi - \beta\|$$

(See Figure 3.2, but also note that ξ is sometimes outside the triangle.)

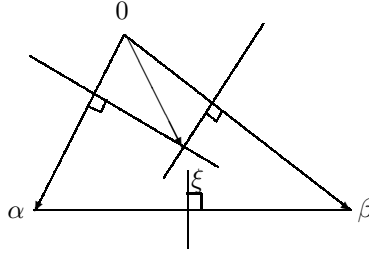


Figure 3.2

It now follows that $\|\xi - \alpha\| = \|\xi - \beta\|$ and thus, by Theorem 3.1, ξ is also a perpendicular bisector of the segment determined by α and β .

The higher dimensional version of a perpendicular bisector of a segment is perpendicular equisector of the convex hull of a set of vectors. The definition depends on a vector μ having properties which correspond to those of the midpoint of a segment.

3.3 Theorem

If $\{\xi_1, \dots, \xi_k\}$ is a linearly independent set in R_n , then there exists a unique $\xi \in [\xi_1, \dots, \xi_k]$ such that

$$\|\xi\| = \|\xi - \xi_i\| \text{ for } i = 1, \dots, k$$

Proof: Rewriting the preceding equation as $(\xi, \xi_i) = \frac{\|\xi_i\|^2}{2}$ for $i = 1, \dots, k$ and letting $\xi = x_1\xi_1 + \dots + x_k\xi_k$, we obtain the matrix equation $GX = B$, where G is the Gram matrix of $\{\xi_1, \dots, \xi_k\}$, $x' = (x_1, \dots, x_k)$, and $B' = (\frac{\|\xi_1\|^2}{2}, \dots, \frac{\|\xi_k\|^2}{2})$.

It follows from Theorem 1.1 that G is nonsingular and hence ξ is uniquely defined.

3.4 Corollary

If $\xi_0, \dots, \xi_k \in R_n$ and $\{\xi_1 - \xi_0, \dots, \xi_k - \xi_0\}$ is a linearly independent set in R_n , then there exists a unique $\mu \in P(\xi_0, \dots, \xi_k)$ such that

$$\|\mu - \xi_0\| = \|\mu - \xi_i\| \text{ for } i = 1, \dots, k$$

Proof: If $\xi \in [\xi_1 - \xi_0, \dots, \xi_k - \xi_0]$ is given as in Theorem 3.3 for the set $\{\xi_1 - \xi_0, \dots, \xi_k - \xi_0\}$, then $\mu = \xi + \xi_0$ satisfies the conditions of this corollary.

The uniqueness of μ follows from the uniqueness of the representation $\mu = \xi_0 + \sum_{i=1}^k x_i(\xi_i - \xi_0)$ in the plane $P(\xi_0, \dots, \xi_k)$ and the uniqueness of ξ in Theorem 3.3. Note that if $\mu = \sum_{i=0}^k a_i \xi_i$, then $\sum_{i=0}^k a_i = 1$. Also note that μ does not always belong to $H(\xi_0, \dots, \xi_k)$, as some of the a_i 's may be negative.

Suppose now that $\xi_0, \dots, \xi_k \in R_n$ and that $\{\xi_1 - \xi_0, \dots, \xi_k - \xi_0\}$ is a linearly independent set in R_n . Let $\mu \in P(\xi_0, \dots, \xi_k)$ be the unique vector given in Corollary 3.4 and note that μ is the analogue of a midpoint for $H(\xi_0, \dots, \xi_k)$. A vector $\xi \in R_n$ is a *perpendicular equisector* of $H(\xi_0, \dots, \xi_k)$ if $(\xi - \mu, \xi_i - \xi_0) = 0$ for $i = 1, \dots, k$. Note that the set of perpendicular equisectors of $H(\xi_0, \dots, \xi_k)$ is the $(n - k)$ -dimensional plane through μ which is perpendicular to $P(\xi_0, \dots, \xi_k)$.

As in Theorem 3.1, we obtain

3.5 Theorem

A vector $\xi \in R_n$ is a perpendicular equisector of $H(\xi_0, \dots, \xi_k)$ if and only if $\|\xi - \xi_0\| = \|\xi - \xi_i\|$ for $i = 1, \dots, k$.

The higher dimensional analogue of Theorem 3.2 is

3.6 Theorem

If $\{\xi_1, \dots, \xi_n\}$ is a linearly independent set in R_n , then there exists a unique perpendicular equisector of the $n + 1$ faces of $H(0, \xi_1, \dots, \xi_n)$.

Proof: From Theorem 3.3, there exists a unique $\xi \in [\xi_1, \dots, \xi_n] = R_n$ such that $\|\xi\| = \|\xi - \xi_i\|$ for $i = 1, \dots, n$. It follows from Theorem 3.5 that ξ is a perpendicular equisector of the faces $H(0, \xi_1, \dots, \hat{\xi}_i, \dots, \xi_n)$ for $i = 1, \dots, n$, where $\hat{\xi}_i$ means that ξ_i is deleted. It also follows from these equations that $\|\xi - \xi_1\| = \|\xi - \xi_j\|$ for $j = 2, \dots, n$, and thus by Theorem 3.5 we see that ξ is also a perpendicular equisector of the last face $H(\xi_1, \dots, \xi_n)$ of $H(0, \xi_1, \dots, \xi_n)$.

Exercises

1. For a linearly independent set $\{\xi_1, \dots, \xi_k\}$ in R_n and for a positive integer m with $m < k$, compare the sets of perpendicular equisectors of $H(0, \xi_1, \dots, \xi_k)$ and of $H(0, \xi_1, \dots, \xi_m)$.

2. For a linearly independent set $\{\alpha, \beta\}$ in R_3 , describe the intersection of the sets of perpendicular equisectors of the sides of the triangle $H(0, \alpha, \beta)$. Generalize this to a linearly independent set, $\{\xi_1, \dots, \xi_k\}$ in R_n .
3. If $\{\xi_1, \dots, \xi_n\}$ is a linearly independent set in R_n , then there exists a unique perpendicular equisector of the $\binom{n+1}{2}$ "edges" of $H(0, \xi_1, \dots, \xi_n)$. Generalize this.
4. Express the unique $\xi \in R_n$ found in Theorem 3.6 in terms of $\xi_1, \dots, \xi_n$ and the Gram determinant g of $\{\xi_1, \dots, \xi_n\}$. Note that this ξ is the unique perpendicular equisector of $H(0, \xi_1, \dots, \xi_n)$ and that it plays an important role in Exercise 3.3.

4 Altitudes

Let $\{\alpha, \beta\}$ be a linearly independent set in R_2 ; consider the triangle $H(0, \alpha, \beta)$. A vector $\xi \in R_2$ is an altitude of $H(0, \alpha, \beta)$ at 0 if $(\xi, \beta - \alpha) = 0$, i.e., ξ belongs to the line L_0 which passes through 0 and is perpendicular to $\beta - \alpha$. A vector $\xi \in R_2$ is an altitude of $H(0, \alpha, \beta)$ at α [respectively, β] if $(\xi, \beta - \alpha) = 0$ [respectively, $(\xi - \beta, \alpha) = 0$], i.e., ξ belongs to the line L_1 [respectively, L_2] which passes through α [respectively, β] and is perpendicular to β [respectively, α]. (See Figure 4.1, but also note that L_1 and L_2 may intersect $H(0, \alpha, \beta)$ at single points.) This figure suggests the following well-known result.

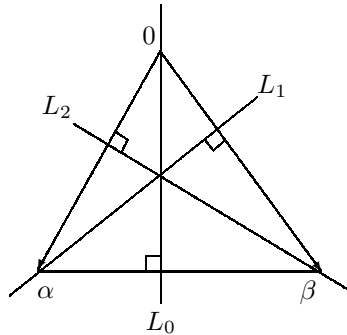


Figure 4.1

4.1 Theorem

The three altitudes of a triangle are concurrent. (That is, there is a unique vector common to the three lines L_0, L_1, L_2 .)

Proof: Since $\{\alpha, \beta\}$ is a linearly independent set in R_2 , the lines L_1 and L_2 are not parallel and hence there exists a unique $\xi \in R_2$ which belongs to $L_1 \cap L_2$. To show that this unique ξ also belongs to L_0 , we show that the

equations $(\xi - \alpha, \beta) = 0$ and $(\xi - \beta, \alpha) = 0$ imply that $(\xi, \beta - \alpha) = 0$. For this, we have

$$\begin{aligned} (\xi, \beta - \alpha) &= (\xi - \alpha + \alpha, \beta - \alpha) \\ &= (\xi - \alpha, \beta) - (\xi - \alpha, \alpha) + (\alpha, \beta - \alpha) \\ &= -(\xi - \beta + \beta - \alpha, \alpha) + (\alpha, \beta - \alpha) \\ &= -(\xi - \beta, \alpha) - (\beta - \alpha, \alpha) + (\alpha, \beta - \alpha) = 0 \end{aligned}$$

We will now use vector algebra to find the common altitude whose existence is guaranteed by the preceding theorem. Since $\{\alpha, \beta\}$ is a linearly independent set in R_2 , we can write $\xi = x\alpha + y\beta$ with $x, y \in R$. The equations for L_1 and L_2 now become

$$([x - 1]\alpha + y\beta, \beta) = 0 \text{ and } (x\alpha + [y - 1]\beta, \alpha) = 0$$

and this leads to the linear system of equations

$$\begin{cases} (\alpha, \alpha)x + (\alpha, \beta)y = (\alpha, \beta) \\ (\beta, \alpha)x + (\beta, \beta)y = (\alpha, \beta) \end{cases}$$

Since the coefficient matrix is the Gram matrix of the linearly independent set, we can solve this system by Cramer's Rule to get

$$x = (\alpha, \beta) \frac{[||\beta||^2 - (\alpha, \beta)]}{g} \text{ and } y = (\alpha, \beta) \frac{[||\alpha||^2 - (\alpha, \beta)]}{g}$$

where g is the Gram determinant of $\{\alpha, \beta\}$. The unique ξ given by Theorem 4.1 is now seen to be

$$\xi = (\alpha, \beta) \frac{[||\beta||^2 - (\alpha, \beta)]\alpha + [||\alpha||^2 - (\alpha, \beta)]\beta}{g}$$

In turning to higher dimensions, we consider a linearly independent set $\{\xi_1, \dots, \xi_n\}$ in R_n . A vector $\xi \in R_n$ is an altitude of $H(0, \xi_1, \dots, \xi_n)$ at 0 if ξ belongs to the line L_0 which passes through 0 and is perpendicular to the plane $P(\xi_1, \dots, \xi_n)$. A vector $\xi \in R_n$ is an altitude of $H(0, \xi_1, \dots, \xi_n)$ at ξ_i ($i = 1, \dots, n$) if $(\xi - \xi_1, \xi_k)$ for $k = 1, \dots, \hat{i}, \dots, n$ ($i = 1, \dots, n$), where $\hat{i}$ means that i is deleted for the set. Therefore ξ is an altitude of $H(0, \xi_1, \dots, \xi_n)$ at ξ_i if and only if ξ belongs to the line L_i which passes through ξ_i and is perpendicular to $\xi_1, \dots, \hat{\xi}_i, \dots, \xi_n$.

It is easy to see that the $n+1$ altitudes of $H(0, \xi_1, \dots, \xi_n)$ are not necessarily concurrent. Consider the example of $\xi_1 = (1, 0, 0)$, $\xi_2 = (0, 0, 1)$, $\xi_3 = (1, 1, 0)$ in R_3 and note that L_2 is the z -axis, while L_3 is given by the equations $x = 1$, $z = 0$. Hence $L_2 \cap L_3$ is empty.

If $i, j \in \{1, \dots, n\}$ and $i \neq j$, then for each $k = 1, \dots, \hat{i}, \dots, \hat{j}, \dots, n$ we have $(\xi, \xi_k) = (\xi_i, \xi_k)$ and $(\xi, \xi_k) = (\xi_j, \xi_k)$ whenever $\xi \in L_i \cap L_j$. Hence a necessary condition for $L_i \cap L_j$ to be nonempty is that

$$(\xi_i, \xi_k) = (\xi_j, \xi_k) \text{ whenever } k = 1, \dots, \hat{i}, \dots, \hat{j}, \dots, n$$

It now follows that a necessary condition for $L_1 \cap \dots \cap L_n$ to be nonempty is that

4.2 Equation

(ξ_i, ξ_j) is independent of i and j with $i \neq j \in \{1, \dots, n\}$.

An example of a linearly independent set in R_3 which satisfies (4.2) but is not an orthogonal set is $\xi_1 = (1, 0, 0)$, $\xi_2 = (1, 0, 1)$, $\xi_3 = (1, 1, 0)$.

4.3 Theorem

The altitudes of $H(0, \xi_1, \dots, \xi_n)$ are concurrent, that is, there is a unique vector common to the $n + 1$ lines $L_0, \dots, L_n$, if and only if (4.2) holds.

Proof: The necessity of (4.2) has already been demonstrated. For the sufficiency, we suppose that $(\xi_i, \xi_j) = b$ whenever $i \neq j \in \{1, \dots, n\}$. It now follows that $\xi \in L_1 \cap \dots \cap L_n$ if and only if $(\xi, \xi_i) = b$ for $i = 1, \dots, n$. Such a ξ obviously satisfies $(\xi, \xi_i - \xi_j) = 0$ whenever $i \neq j$ and hence belongs to L_0 . We must now show that there is a *unique* $\xi \in R_n$ which satisfies $(\xi, \xi_i) = b$ for $i = 1, \dots, n$. Writing $\xi = \sum_{i=1}^n x_i \xi_i$ with $x_i \in R$ yields the linear system of equations $GX = B$, where G is the Gram matrix of $\{\xi_1, \dots, \xi_n\}$, $X' = (x_1, \dots, x_n)$, and $B' = (b, \dots, b)$. Once again, G is non-singular by Theorem 1.1 and hence ξ is unique.

Note that the unique $\xi \in R_n$ given by the preceding theorem can be found by Cramer's Rule as in the case $n = 2$ and is

$$\xi = b \frac{(\sum_{i=1}^n g_i \xi_i)}{g}$$

where g is the Gram determinant of $\{\xi_1, \dots, \xi_n\}$ and g_i is the determinant of the matrix obtained from G by replacing the i -th column of G by the column with all entries equal to one.

Here is another interesting problem related to altitudes. Let $\{\xi_1, \dots, \xi_n\}$ be a linearly independent set in R_n , and find the vector δ of smallest norm in $P(\xi_1, \dots, \xi_n)$. From geometric considerations, it follows that δ is the unique vector in $L_0 \cap P(\xi_1, \dots, \xi_n)$. In the preceding notation, Griffin [4] has shown that

$$\delta = \sum_{i=1}^n g_i \frac{\xi_i}{g} \text{ and } \|\delta\|^2 = g / \left(\sum_{i=1}^n g_i \right)$$

The norm of unique vector $\delta_i \in R_n$ for which $\xi_i + \delta_i \in L_i \cap [\xi_1, \dots, \hat{\xi}_i, \dots, \xi_n]$ is obtained by finding the distance from ξ_i to the hyperplane $[\xi_1, \dots, \hat{\xi}_i, \dots, \xi_n]$. In this way, it is easily shown that

$$\|\delta_i\|^2 = \frac{g}{\hat{g}_i}$$

where g_i is the Gram determinant of $\{\xi_1, \dots, \hat{\xi}_i, \dots, \xi_n\}$.

Exercises

1. Prove the well-known theorem of Euclid that the altitudes of $H(0, \alpha, \beta)$ are the perpendicular bisectors of $H(\alpha + \beta, \beta - \alpha, \alpha - \beta)$ in R_2 . Theorem 4.1 now follows from Theorem 3.2.
2. Can Exercise 4.1 be generalized if (4.2) holds?
3. Define altitudes of $H(0, \xi_1, \dots, \xi_k)$ where $\{\xi_1, \dots, \xi_k\}$ is a linearly independent set in R_n . The $k+1$ sets of altitudes $L_0, \dots, L_k$ will be $(n-k+1)$ -dimensional planes in R_n . What are necessary conditions for $L_1 \cap \dots \cap L_k$ to be nonempty? Do these conditions imply that $L_0 \cap \dots \cap L_k$ is nonempty? Describe $L_0 \cap \dots \cap L_k$ and $L_0 \cap \dots \cap L_k \cap [\xi_1, \dots, \xi_k]$.
4. Use the Cauchy-Schwartz inequality to justify the “geometric considerations” which imply that $\delta \in L_0$. That is, show that if $\delta \in L_0 \cap P(\xi_1, \dots, \xi_n)$ then $\|\delta\| \leq \|\alpha\|$ whenever $\alpha \in P(\xi_1, \dots, \xi_n)$.
5. Find the formula for the distance from a point (or vector) to a hyperplane in R_n . Find the formula for the distance from a point (or vector) to a k -dimensional plane in R_n .

5 Centroids and Medians

Let $\{\alpha, \beta\}$ be a linearly independent set in R_2 . The *centroid of the triangle* $H(0, \alpha, \beta)$ is $\gamma = \frac{(\alpha + \beta)}{3}$ (Figure 5.1 (a)), which is, in fact, the point of intersection of the three medians of $H(0, \alpha, \beta)$. In order to study properties of the centroid, we translate the origin to γ (Figure 5.1 (b)). Thus we let

$$\eta_1 = -\gamma, \eta_2 = \alpha - \gamma, \text{ and } \eta_3 = \beta - \gamma$$

and note that $\{\eta_1, \eta_2\}$ is a linearly independent set in R_2 , and $\eta_1 + \eta_2 + \eta_3 = 0$.

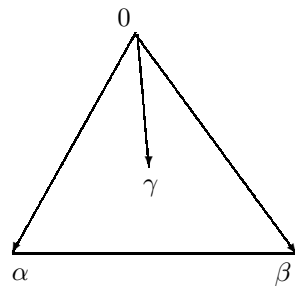


Figure 5.1 (a)

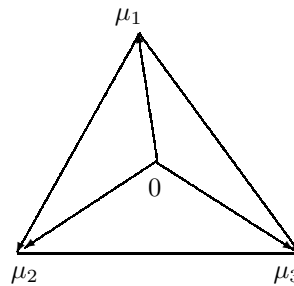


Figure 5.1 (b)

We can now prove a “least squares” theorem.

5.1 Theorem

Let $\eta_1, \eta_2, \eta_3 \in R_2$ with $\{\eta_1, \eta_2\}$ a linearly independent set, and $\eta_1 + \eta_2 + \eta_3 = 0$. Then

$$\|\eta_1 - \xi\|^2 + \|\eta_2 - \xi\|^2 + \|\eta_3 - \xi\|^2$$

is minimal over $\xi \in R_2$ if and only if $\xi = 0$.

Proof: Since $\eta_1 + \eta_2 + \eta_3 = 0$, we see that

$$(\eta_1, \xi) + (\eta_2, \xi) + (\eta_3, \xi) = 0 \text{ for all } \xi \in R_2$$

and hence

$$\begin{aligned} \sum_{i=1}^3 \|\eta_i - \xi\|^2 &= \sum_{i=1}^3 \|\eta_i\|^2 - 2 \sum_{i=1}^3 (\eta_i, \xi) + 3\|\xi\|^2 \\ &= \|\eta_1\|^2 + \|\eta_2\|^2 + \|\eta_3\|^2 + 3\|\xi\|^2 \end{aligned}$$

This is clearly minimal if and only if $\xi = 0$.

5.2 Theorem

If $\eta_1, \eta_2, \eta_3 \in R_2$ as in Theorem 5.1, then the areas of the triangles $H(0, \eta_1, \eta_2)$, $H(0, \eta_1, \eta_3)$, and $H(0, \eta_2, \eta_3)$ are equal.

Proof: It suffices by (2.3) to show that the corresponding Gram determinants are equal, i.e., in the obvious notation, that $g(\eta_1, \eta_2) = g(\eta_1, \eta_3) = g(\eta_2, \eta_3)$. To show that $g(\eta_1, \eta_2) = g(\eta_2, \eta_3)$, we replace η_1 by $-(\eta_2 + \eta_3)$, and calculate as follows.

$$\begin{aligned} g(\eta_1, \eta_2) &= \begin{vmatrix} (\eta_1, \eta_1) & -(\eta_1, \eta_2) \\ -(\eta_2, \eta_1) & (\eta_2, \eta_2) \end{vmatrix} \\ &= \begin{vmatrix} (\eta_2 + \eta_3, \eta_2 + \eta_3) & -(\eta_2 + \eta_3, \eta_2) \\ -(\eta_2, \eta_2 + \eta_3) & (\eta_2, \eta_2) \end{vmatrix} \\ &= \begin{vmatrix} (\eta_2, \eta_2) + 2(\eta_2, \eta_3) + (\eta_3, \eta_3) & -(\eta_2, \eta_2) - (\eta_3, \eta_2) \\ -(\eta_2, \eta_2) - (\eta_2, \eta_3) & (\eta_2, \eta_2) \end{vmatrix} \end{aligned}$$

Adding the second row to the first row gives

$$g(\eta_1, \eta_2) = \begin{vmatrix} (\eta_2, \eta_3) + (\eta_3, \eta_3) & -(\eta_3, \eta_2) \\ -(\eta_2, \eta_2) - (\eta_2, \eta_3) & (\eta_2, \eta_2) \end{vmatrix}$$

Adding the second column to the first column now gives

$$\begin{aligned} g(\eta_1, \eta_2) &= \begin{vmatrix} (\eta_3, \eta_3) & -(\eta_3, \eta_2) \\ -(\eta_2, \eta_3) & (\eta_2, \eta_2) \end{vmatrix} \\ &= g(\eta_2, \eta_3) \end{aligned}$$

The fact that $g(\eta_1, \eta_3) = g(\eta_2, \eta_3)$ is now obvious.

We now suppose that $\{\xi_1, \dots, \xi_k\}$ is a linearly independent set in R_n . The *centroid* of $H(0, \xi_1, \dots, \xi_k)$ is defined to be the vector $\gamma = \frac{(\xi_1 + \dots + \xi_k)}{(k+1)}$. Translation of the origin to γ as in the case $n = 2$ leads to the study of sets of vectors $\eta_1, \dots, \eta_{k+1} \in R_n$, for which $\{\eta_1, \dots, \eta_k\}$ is a linearly independent set and $\eta_1 + \dots + \eta_{k+1} = 0$.

The following two higher dimensional analogues of Theorems 5.1 and 5.2 have been proven by Beck [2].

5.3 Theorem

Let $\eta_1, \dots, \eta_{k+1} \in R_n$ with $\{\eta_1, \dots, \eta_k\}$ a linearly independent set and $\eta_1 + \dots + \eta_{k+1} = 0$. Then

$$\|\eta_1 - \xi\|^2 + \dots + \|\eta_{k+1} - \xi\|^2$$

is minimal over $\xi \in R_n$ if and only if $\xi = 0$.

5.4 Theorem

If $\eta_1, \dots, \eta_{k+1} \in R_n$ are as in Theorem 5.3, then the k -dimensional volume measures of $H(0, \eta_1, \dots, \hat{\eta}_i, \dots, \eta_{k+1})$ are independent of $i = 1, \dots, k+1$.

Exercises

1. Illustrate and prove Theorems 5.3 and 5.4 in the case $n = k = 3$.
2. Prove Theorems 5.3 and 5.4.
3. Find the distances from 0 to the faces of the triangle $H(\eta_1, \eta_2, \eta_3)$ in Figure 5.1(b). Use these distances to give another proof of Theorem 5.2.
4. Prove Theorem 5.4 by generalizing the idea in Exercise 5.3.
5. Define the three sets of medians of a triangle in R_2 via vector algebra. Show that the medians of a triangle in R_2 are concurrent. Generalize this.
6. If $\{\xi_1, \dots, \xi_n\}$ is a linearly independent set in R_n and M_0 denotes the medians of $H(0, \xi_1, \dots, \xi_n)$ at 0, find δ and $\|\delta\|$, where $\delta \in M_0 \cap P(\xi_1, \dots, \xi_n)$.

6 Angle Equisectors

Let $\alpha, \beta, \gamma \in R_2$ with $\{\alpha, \beta\}$ a linearly independent set. A vector $\xi \in R_2$ is an *angle bisector* of α and β at γ if $\left(\xi - \gamma, \frac{\alpha}{\|\alpha\|}\right) = \left(\xi - \gamma, \frac{\beta}{\|\beta\|}\right)$ i.e., the lengths of the perpendicular projections of $\xi - \gamma$ onto $[\alpha]$ and $[\beta]$ are equal (Figure 6.1).

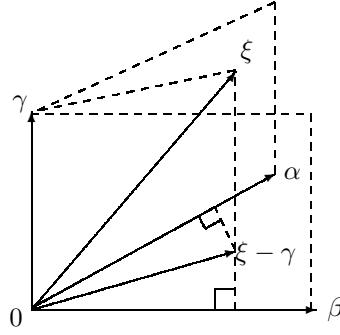


Figure 6.1

The set of all angle bisectors of α and β at γ is the line which passes through γ and is perpendicular to $\left(\frac{\alpha}{\|\alpha\|}\right) - \left(\frac{\beta}{\|\beta\|}\right)$. Note that

$$\left(\frac{\alpha}{\|\alpha\|} - \frac{\beta}{\|\beta\|}, \frac{\alpha}{\|\alpha\|} + \frac{\beta}{\|\beta\|}\right) = \frac{(\alpha, \alpha)}{\|\alpha\|^2} - \frac{(\beta, \beta)}{\|\beta\|^2} = 0$$

and hence that ξ is an angle bisector of α and β at γ if and only if

6.1 Equation

$\xi = \gamma + t\left(\frac{\alpha}{\|\alpha\|} + \frac{\beta}{\|\beta\|}\right)$ for some $t \in R$.

6.2 Theorem

If ξ is an angle bisector of α and β at 0, and P_1 and P_2 denote the perpendicular projections of R_2 onto $[\alpha]$ and $[\beta]$, respectively, then $\|\xi - P_1\xi\| = \|\xi - P_2\xi\|$.

Proof: For any $\xi \in R_2$, we have $P_1\xi = \left(\xi, \frac{\alpha}{\|\alpha\|}\right) \frac{\alpha}{\|\alpha\|}$ and hence $\|P_1\xi\| = \left(\xi, \frac{\alpha}{\|\alpha\|}\right)$ and $(\xi, P_1\xi) = \left(\xi, \frac{\alpha}{\|\alpha\|}\right)^2$. It now follows that

$$\|\xi - P_1\xi\|^2 = \|\xi\|^2 - 2(\xi, P_1\xi) + \|P_1\xi\|^2 = \|\xi\|^2 - (\xi, \alpha/\|\alpha\|)^2$$

If ξ is an angle bisector of α and β at 0, then we clearly have $\|\xi - P_1\xi\|^2 = \|\xi - P_2\xi\|^2$. The preceding theorem states that if ξ is an angle bisector of α and β at 0, then P is equidistant from the sides of the angle determined by α and β . The following result is a consequence of the definitions of angle bisector and altitude.

6.3 Theorem

A vector $\xi \in R_2$ is an angle bisector of α and β at 0, if and only if is an altitude of $H\left(0, \frac{\alpha}{\|\alpha\|}, \frac{\beta}{\|\beta\|}\right)$ at 0.

We continue to assume that $\{\alpha, \beta\}$ is a linearly independent set in R_2 . Let B_0 denote the set of angle bisectors of α and β at 0. We would like to find the unique vector $\mu \in R_2$ which belongs to $B_0 \cap H(\alpha, \beta)$, i.e., the angle bisector of α and β at 0, which belongs to the segment determined by α and β . It follows from (6.1) and the definition of $H(\alpha, \beta)$ that

$$\mu = t \left(\frac{\alpha}{\|\alpha\|} + \frac{\beta}{\|\beta\|} \right) = s\alpha + (1-s)\beta \text{ for some } t \in R \text{ and } s \in [0, 1]$$

The linear independence of the set $\{\alpha, \beta\}$ implies that

$$t \left(\frac{1}{\|\alpha\|} + \frac{1}{\|\beta\|} \right) = 1 \text{ or } t = \frac{\|\alpha\| \|\beta\|}{\|\alpha\| + \|\beta\|}$$

We therefore see that

$$\mu = \frac{(\|\beta\|\alpha + \|\alpha\|\beta)}{(\|\alpha\| + \|\beta\|)}$$

The norm of μ can be computed by observing that

$$(\|\beta\|\alpha + \|\alpha\|\beta, \|\beta\|\alpha + \|\alpha\|\beta) = 2\|\alpha\| \|\beta\| [\|\alpha\| \|\beta\| + (\alpha, \beta)]$$

and hence giving

$$\|\mu\|^2 = 2\|\alpha\| \|\beta\| \frac{[\|\alpha\| \|\beta\| + (\alpha, \beta)]}{(\|\alpha\| + \|\beta\|)^2}$$

We now consider the intersection of the sets of angle bisectors of a triangle. Let $\{\alpha, \beta\}$ be a linearly independent set in R_2 and consider the triangle $H(0, \alpha, \beta)$. A vector $\xi \in R_2$ is an angle bisector of $H(0, \alpha, \beta)$ at 0 if ξ is an angle bisector of α and β at 0. A vector $\xi \in R_2$ is an angle bisector of $H(0, \alpha, \beta)$ at α if ξ is an angle bisector of $-\alpha$ and $\beta - \alpha$ at α . A vector $\xi \in R_2$ is an angle bisector of $H(0, \alpha, \beta)$ at β if ξ is an angle bisector of $-\beta$ and $\alpha - \beta$ at β . We therefore have B_0, B_1, B_2 , the three lines of angle bisectors of $H(0, \alpha, \beta)$, as in Figure 6.2.

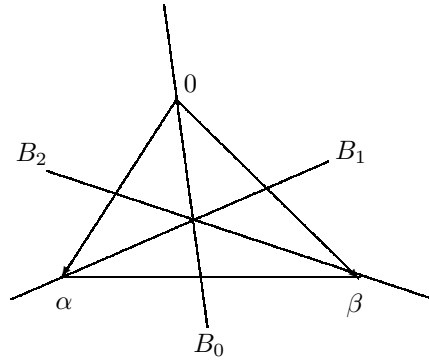


Figure 6.2

6.4 Theorem

The angle bisectors of a triangle are concurrent. (That is, there is a unique vector common to the three lines B_0, B_1, B_2 .)

Proof: Since $\{\alpha, \beta\}$ is a linearly independent set, the lines B_0 and B_1 are not parallel and hence, using the form (6.1), there exists a unique $\xi \in R_2$ such that

$$\xi = t \left(\frac{\alpha}{\|\alpha\|} + \frac{\beta}{\|\beta\|} \right) = \alpha - s \left(\frac{\alpha}{\|\alpha\|} + \frac{\alpha - \beta}{\|\alpha - \beta\|} \right) \text{ for some } s, t \in R$$

Therefore

$$\frac{t}{\|\alpha\|} \alpha + \frac{t}{\|\beta\|} \beta = \left(1 - \frac{s}{\|\alpha\|} - \frac{s}{\|\alpha - \beta\|} \right) \alpha + \frac{s}{\|\alpha - \beta\|} \beta$$

and the linear independence of the set $\{\alpha, \beta\}$ implies that

$$\frac{t}{\|\alpha\|} = 1 - \frac{s}{\|\alpha\|} - \frac{s}{\|\alpha - \beta\|} \text{ and } \frac{t}{\|\beta\|} = \frac{s}{\|\alpha - \beta\|}$$

Substituting $s = \frac{\|\alpha - \beta\| t}{\|\beta\|}$ into the first equation and solving for t gives

$$t = \frac{\|\alpha\| \|\beta\|}{(\|\alpha\| + \|\beta\| + \|\alpha - \beta\|)}$$

The unique $\xi \in R_2$ in $B_0 \cap B_1$ is therefore

6.5 Equation

$$\xi = \frac{(\|\beta\|\alpha + \|\alpha\|\beta)}{(\|\alpha\| + \|\beta\| + \|\alpha - \beta\|)}$$

Interchanging α and β in the preceding argument shows that $\xi \in B_0 \cap B_2$ and hence that $\xi \in B_0 \cap B_1 \cap B_2$, as desired.

Note that $\xi \in H(0, \alpha, \beta)$ and (as with μ) that

$$\|\xi\|^2 = 2 \|\alpha\| \|\beta\| \frac{[\|\alpha\| \|\beta\| + (\alpha, \beta)]}{(\|\alpha\| + \|\beta\| + \|\alpha - \beta\|)^2}$$

In higher dimensions, we consider $\xi_0, \dots, \xi_n \in R_n$ with $\{\xi_1, \dots, \xi_n\}$ a linearly independent set in R_n . A vector $\xi \in R_n$ is an *angle equisector* of $\{\xi_1, \dots, \xi_n\}$ at ξ_0 if $\left(\xi - \xi_0, \frac{\xi_i}{\|\xi_i\|} \right) = \left(\xi - \xi_0, \frac{\xi_j}{\|\xi_j\|} \right)$ for $i, j = 1, \dots, n$, i.e., the lengths of the orthogonal projections of $\xi - \xi_0$ onto $[\xi_i]$ for $i = 1, \dots, n$ are equal. The set of all angle equisectors of $\{\xi_1, \dots, \xi_n\}$ at ξ_0 is a line which passes through ξ_0 and is perpendicular to $\left(\frac{\xi_i}{\|\xi_i\|} \right) - \left(\frac{\xi_j}{\|\xi_j\|} \right)$ for $i, j = 1, \dots, n$. As in the case $n = 2$, we obtain

6.6 Theorem

If ξ is an angle equisector of $\{\xi_1, \dots, \xi_n\}$ at 0, and P_i denotes the orthogonal projection of R_n onto $[\xi_i]$, then $\|\xi - P_i \xi\| = \|\xi - P_j \xi\|$ for $i, j = 1, \dots, n$.

For a linearly independent set $\{n\}$ in R_n $\{\xi_1, \dots, \xi_n\}$ at R_n , let B_0 denote the set of angle equisectors of $\{\xi_1, \dots, \xi_n\}$ at 0. As in the case $n = 2$, we would like to find the unique $\mu \in R_n$ which is in $B_0 \cap H(\xi_1, \dots, \xi_n)$. To do this, it is convenient to find a direction vector for the line B_0 so that we have a form similar to (6.1). One method for finding a direction vector for B_0 is to note, as in Theorem 6.3, that $\xi \in B_0$ if and only if ξ is an altitude of $H\left(0, \frac{\xi_1}{\|\xi_1\|}, \dots, \frac{\xi_n}{\|\xi_n\|}\right)$ at 0. Hence the vector δ (or a nonzero multiple of δ) given in Section 4 with ξ_i replaced by $\frac{\xi_i}{\|\xi_i\|}$ is a direction vector for B_0 . Alternatively, one can find a direction vector for B_0 by noting that $\nu \in B_0$ if $(\nu, \xi_i) = \|\xi_i\|$ for $i = 1, \dots, n$. Letting $\nu = \sum_{i=1}^n x_i \xi_i$, we obtain the linear system of equations $GX = B$, where G is the Gram matrix of $\{\xi_1, \dots, \xi_n\}$, $X' = (x_1, \dots, x_n)$ and $B' = (\|\xi_1\|, \dots, \|\xi_n\|)$. From Cramer's Rule, we now conclude that the unique ν satisfying $(\nu, \xi_i) = \|\xi_i\|$ for $i = 1, \dots, n$ is

$$\nu = \sum_{i=1}^n h_i \xi_i / g$$

where g is the Gram determinant of $\{\xi_1, \dots, \xi_n\}$, and h_i is the determinant of the matrix obtained by replacing the i -th column of the Gram matrix of $\{\xi_1, \dots, \xi_n\}$ by $(\|\xi_1\|, \dots, \|\xi_n\|)'$. We now conclude that $\xi \in B_0$ if and only if

6.7 Equation

$$\xi = t \sum_{i=1}^n h_i \xi_i / g \text{ for some } t \in R$$

Hence $\mu \in B_0 \cap H(\xi_1, \dots, \xi_n)$ if and only if

$$\mu = t \sum_{i=1}^n h_i \xi_i / g = \sum_{i=1}^n a_i \xi_i \text{ for some } t \in R \text{ and } a_i \geq 0 \text{ with } \sum_{i=1}^n a_i = 1$$

The linear independence of the set $\{\xi_1, \dots, \xi_n\}$ implies that

$$t \sum_{i=1}^n h_i / g = 1 \text{ or } t = g / \sum_{i=1}^n h_i$$

We therefore see that

$$\mu = \sum_{i=1}^n h_i \xi_i / \sum_{i=1}^n h_i$$

In considering the concurrency of the sets of angle equisectors of the convex hull of a set of vectors, we encounter problems similar to those for altitudes.

Let $\{\xi_1, \dots, \xi_n\}$ be a linearly independent set in R_n . A vector $\xi \in R_n$ is angle equisector of $H(0, \xi_1, \dots, \xi_n)$ at 0 if ξ is an angle equisector of $\{\xi_1, \dots, \xi_n\}$ at 0. A vector $\xi \in R_n$ is an angle equisector of $H(0, \xi_1, \dots, \xi_n)$ at $\xi_i (i = 1, \dots, n)$ if ξ is an angle equisector of $\{-\xi_i, \xi_1 - \xi_i, \dots, (\xi_i - \xi_i), \dots, \xi_n - \xi_i\}$ at ξ_i . In the obvious notation, we have $B_0, \dots, B_n$, the $n + 1$ lines of angle equisectors of $H(0, \xi_1, \dots, \xi_n)$.

Note that if $\xi \in B_0$, then $\left(\xi, \frac{\xi_i}{\|\xi_i\|}\right) = \left(\xi, \frac{\xi_j}{\|\xi_j\|}\right)$ for all $i, j = 1, \dots, n$, and hence $\left(\xi, \frac{\xi_i}{\|\xi_i\|}\right)$ is independent of $i = 1, \dots, n$. Therefore if $\xi \in B_0 \cap B_i$ for some $i = 1, \dots, n$, then

$$\left(\xi, \frac{\xi_1}{\|\xi_1\|}\right) = \dots = \left(\xi, \frac{\xi_n}{\|\xi_n\|}\right)$$

(say, $= d$), and

$$\left(\xi - \xi_i, \frac{-\xi_i}{\|\xi_i\|}\right) = \left(\xi - \xi_i, \frac{(\xi_j - \xi_i)}{\|\xi_i - \xi_j\|}\right) \text{ for } j = 1, \dots, \hat{i}, \dots, n$$

It now follows from these equations that

$$\begin{aligned} -\left(\xi, \frac{\xi_i}{\|\xi_i\|}\right) + \|\xi_i\| &= \frac{[(\xi, \xi_j) - (\xi, \xi_i) + (\xi_i, \xi_i - \xi_j)]}{\|\xi_i - \xi_j\|}, -d + \|\xi_i\| \\ &= \frac{\|\xi_j\|d - \|\xi_i\|d + (\xi_i, \xi_i - \xi_j)}{\|\xi_i - \xi_j\|} \end{aligned}$$

One can algebraically solve for d to get

$$d = \frac{[(\xi_i, \xi_i - \xi_j) - \|\xi_i\| \|\xi_i - \xi_j\|]}{(\|\xi_i\| - \|\xi_j\| - \|\xi_i - \xi_j\|)}$$

We therefore see that a necessary condition for $B_0 \cap B_i$ to be nonempty is that the right-hand side of the preceding equation for d is independent of $j = 1, \dots, \hat{i}, \dots, n$. It now follows that a necessary condition for $B_0 \cap \dots \cap B_n$ to be nonempty is that

6.8 Equation

$$\frac{[(\xi_i, \xi_i - \xi_j) - \|\xi_i\| \|\xi_i - \xi_j\|]}{(\|\xi_i\| - \|\xi_j\| - \|\xi_i - \xi_j\|)}$$

is independent of $i \neq j \in \{1, \dots, n\}$.

6.9 Theorem

The angle equisectors of $H(0, \xi_1, \dots, \xi_n)$ are concurrent; that is, there is a unique vector common to the $n + 1$ lines $B_0, \dots, B_n$, if and only if (6.8) holds.

Proof: The necessity of (6.8) has already been demonstrated. For the sufficiency, we suppose that (6.8) holds and we denote the common value by d . As usual, there is a unique $\xi \in R_n$ for which $\left(\xi, \frac{\xi_1}{\|\xi_1\|}\right) = \cdots = \left(\xi, \frac{\xi_n}{\|\xi_n\|}\right) = d$. Thus $\xi \in B_0$ and the preceding calculation shows that $\xi \in B_1$ for each $i = 1, \dots, n$. The uniqueness of this follows from the preceding calculations.

Note that the unique $\xi \in R_n$ given by the preceding theorem is

$$\xi = d\nu = d\left(\sum_{i=1}^n h_i \xi_i\right)/g$$

where g is the Gram determinant of $\{\xi_1, \dots, \xi_n\}$ and h_i is the determinant of the matrix obtained by replacing the i -th column of the Gram matrix of $\{\xi_1, \dots, \xi_n\}$ by $(\|\xi_1\|, \dots, \|\xi_n\|)'$.

Exercises

1. Show that $\xi = \frac{(\|\beta\|\alpha + \|\alpha\|\beta)}{(\|\alpha\| + \|\beta\| + \|\alpha - \beta\|)}$ belongs to $B_0 \cap B_1 \cap B_2$ as in Theorem 6.4 by showing that it satisfies the angle bisector equations of $H(0, \alpha, \beta)$:

$$\begin{cases} \left(\xi, \frac{\alpha}{\|\alpha\|}\right) &= \left(\xi, \frac{\beta}{\|\beta\|}\right) \\ \left(\xi - \alpha, \frac{\alpha}{\|\alpha\|}\right) &= \left(\xi - \alpha, \frac{(\alpha - \beta)}{\|\alpha - \beta\|}\right) \\ \left(\xi - \beta, \frac{\beta}{\|\beta\|}\right) &= \left(\xi - \beta, \frac{(\beta - \alpha)}{\|\alpha - \beta\|}\right) \end{cases}$$

2. Show that any pair of the angle bisector equations of $H(0, \alpha, \beta)$ given in Exercise 6.1 imply the third equation.
3. Prove the following improved version of Theorem 6.2. A vector $\xi \in R_2$ is an angle bisector of α and β at γ if and only if $\|\xi - \gamma - P_1(\xi - \gamma)\| = \|\xi - \gamma - P_2(\xi - \gamma)\|$ and $\xi = \gamma + t(s\alpha + (1 - s)\beta)$ for some $t \in R$ and $s \in [0, 1]$. The latter condition says that $\xi - \gamma$ is a scalar multiple of a vector belonging to the segment $H(\alpha, \beta)$.
4. Use Exercise 6.3 to show Exercise 6.2 by noting that $\xi \in B_0 \cap B_1 \cap B_2$ if and only if $\xi \in H(0, \alpha, \beta)$ and

$$\begin{cases} B_0 : & \|\xi - P_1\xi\| = \|\xi - P_2\xi\| \\ B_1 : & \|\xi - P_1\xi\| = \|\xi - \alpha - P_3(\xi - \alpha)\| \\ B_2 : & \|\xi - P_2\xi\| = \|\xi - \beta - P_3(\xi - \beta)\| \end{cases}$$

where P_3 denotes the orthogonal projection of R_2 onto $[\alpha - \beta]$. Interpret this result geometrically.

5. Generalize the idea in Exercise 6.4 to higher dimensions.

6. Let ξ be the unique vector in R_2 given by Theorem 6.4. Show that ξ is equidistant from the three sides of $H(0, \alpha, \beta)$ and then find this distance r as a function of α and β . Relate this to Exercise 6.4. Generalize this to higher dimensions.
7. For a linearly independent set $\{\alpha, \beta\}$ in R_3 , define angle bisectors of $H(0, \alpha, \beta)$ by the equations in Exercise 6.1. Show that $B_0 \cap B_1 \cap B_2$ is the line through (defined in Exercise 6.1) which is perpendicular to $[\alpha, \beta]$. Generalize this to $H(0, \xi_1, \dots, \xi_k)$, where $\{\xi_1, \dots, \xi_k\}$ is a linearly independent set in R_n .
8. Show that condition (6.8) is symmetric in i and j , and hence conclude that the phrase “independent of $i \neq j \in \{1, \dots, n\}$ ” can be replaced by “independent of $1 \leq i < j \leq n$ ”. We now see that condition (6.8) is vacuous if $n = 2$ and this is consistent with Theorem 6.4.
9. Investigate condition (6.8) in the case that $n = 3$ and $\|\xi_1\| = \|\xi_2\| = \|\xi_3\|$. More generally, define what it means for $H(0, \xi_1, \dots, \xi_n)$ to be isosceles (at 0) and investigate condition (6.8) in this setting.

7 Problems

In this section, we state some additional exercises, many of which relate topics from more than one of the preceding sections and may lead to more advanced research problems. A geometrical understanding of the two dimensional problem is essential before posing generalizations to higher dimensions.

Exercises

1. Let ξ denote the unique common angle bisector of $H(0, \alpha, \beta)$ given in Theorem 6.4. Use the idea of Exercise 6.6 to show that

$$\frac{A_0}{\|\alpha - \beta\|} = \frac{A_1}{\|\alpha\|} = \frac{A_2}{\|\beta\|}$$

where A_0, A_1, A_2 are the areas of $H(\alpha, \beta, \xi), H(0, \alpha, \xi), H(0, \beta, \xi)$, respectively. What does this say about the Gram determinants $g(\alpha, \xi), g(\beta, \xi)$, and $g(\xi - \alpha, \xi - \beta)$?

2. Let ξ denote the unique common angle bisector of $H(0, \alpha, \beta)$ given in Theorem 6.4, and let δ denote the unique altitude vector of $H(0, \alpha, \beta)$ at 0 which belongs to $P(\alpha, \beta)$ given in Section 4. Show by geometry and by the formula in Section 4, that $\|\delta\|^2 = \frac{g}{\|\alpha - \beta\|}^2$. Then show that $\frac{r}{\|\delta\|} = \frac{\|\alpha - \beta\|}{k}$, where r is the distance in Exercise 6.6 and $k = \|\alpha\| + \|\beta\| + \|\alpha - \beta\|$. Find similar formulas for the vectors δ_1 and δ_2 defined in Section 4. Show that

$$\frac{1}{r} = \frac{1}{\|\delta\|} + \frac{1}{\|\delta_1\|} + \frac{1}{\|\delta_2\|}$$

3. For any $\gamma \in H(0, \alpha, \beta)$, let ℓ, ℓ_1, ℓ_2 denote the distances from γ to $P(\alpha, \beta)$, $[\beta]$, $[\alpha]$, respectively. Prove that

$$\frac{\ell}{\|\delta\|} + \frac{\ell_1}{\|\delta_1\|} + \frac{\ell_2}{\|\delta_2\|} = 1$$

where $\delta, \delta_1, \delta_2$ are the altitudes of $H(0, \alpha, \beta)$ defined in Section 4. What happens if $\gamma \notin H(0, \alpha, \beta)$?

4. Try to generalize Exercises 7.1 - 7.3 to three dimensions and then to higher dimensions. It will be useful to develop a theory to find a vector in R_3 which is equidistant from the faces of $H(0, \xi_1, \xi_2, \xi_3)$ and then generalize this to higher dimensions.
5. Prove that conditions (4.2) and (6.8) are equivalent if $H(0, \xi_1, \dots, \xi_n)$ is isosceles (at 0). (See Exercise 6.9.) Investigate implications and relationships between conditions (4.2) and (6.8). In particular, try to find situations in which $H(0, \xi_1, \dots, \xi_n)$ has concurrent altitudes if and only if it has concurrent angle equisectors. Also prove that the unique perpendicular equisector of $H(0, \xi_1, \dots, \xi_n)$ (see Exercise 3.4) is an angle equisector of $(\xi_1, \dots, \xi_n)$ at 0 if $H(0, \xi_1, \dots, \xi_n)$ is isosceles.
6. Prove the following result in the setting of Exercise 7.2. If $\delta \in H(\alpha, \beta)$ and $\delta \neq \alpha, \beta$, then the set of altitudes of $H(0, \alpha, \beta)$ at 0 is equal to the set of angle bisectors of $H(\delta, \alpha + \delta_1, \beta + \delta_2)$ at δ . What happens if $\delta \notin H(\alpha, \beta)$? Generalize this to higher dimensions.
7. Find relationships between perpendicular equisectors, altitudes, medians, and angle equisectors similar to those in Exercises 7.6 and 4.1.
8. Let μ denote the unique angle bisector of α and β at 0 which belongs to $H(\alpha, \beta)$. Prove that

$$\frac{\|\mu - \alpha\|}{\|\mu - \beta\|} = \frac{\|\alpha\|}{\|\beta\|}$$

Try to generalize this result to higher dimensions.

9. Let ξ denote the unique common angle bisector of $H(0, \alpha, \beta)$ given in Theorem 6.4, and let ψ denote the unique common perpendicular bisector of $H(0, \alpha, \beta)$ given in Theorem 3.2. Let r denote the distance from ξ to the sides of $H(0, \alpha, \beta)$ as in Exercise 6.6 and let R denote the distance from ψ to the corners of $H(0, \alpha, \beta)$ as in Theorem 3.1. Prove Euler's Theorem:

$$\|\xi - \psi\|^2 = R(R - r)$$

Generalize this result to higher dimensions.

10. Consider a triangle $H(0, \alpha, \beta)$ in R_2 with $(\alpha, \beta) = 0$. Let δ be the unique altitude vector of $H(0, \alpha, \beta)$ at 0 which belongs to $P(\alpha, \beta)$ and hence to $H(\alpha, \beta)$. Let P_1, P_2, P_3 denote the orthogonal projections of R_2 onto

$[\alpha], [\beta], [\alpha - \beta]$, respectively. Define two sequences of vectors $\{\alpha_n\}$ and $\{\mu_n\}$ according to the scheme:

$$\alpha_1 = P_1\delta, \quad \mu_1 = \delta + P_3\alpha_1, \quad \alpha_2 = P_1\mu_1, \quad \mu_2 = \delta + P_3\alpha_2, \quad \dots$$

Note that $\alpha_n \in [\alpha]$, $\mu_n \in H(\alpha, \beta)$, and μ_n is the unique altitude vector of $H(\alpha_n, \mu_{n-1}, \alpha)$ at α_n which belongs to $H(\alpha, \beta)$. Another pair of sequences of vectors $\{\beta_n\}$ and $\{\nu_n\}$ is defined by replacing P_l by P_2 in the preceding scheme. Hence $\beta_n \in [\beta]$ and $\nu_n \in H(\alpha, \beta)$. Use induction to prove that

$$\frac{\|\alpha_n - \alpha\|}{\|\beta_n - \beta\|} = \left(\frac{\|\alpha\|}{\|\beta\|} \right) \quad \text{and} \quad \frac{\|\alpha_n - \mu_n\|}{\|\beta_n - \nu_n\|} = \left(\frac{\|\alpha\|}{\|\beta\|} \right)^{2n}$$

11. Consider an equilateral triangle $H(0, \alpha, \beta)$ in R_2 , and let ψ and R be as in Exercise 7.9. Show that $\|\psi\| = \|\psi - \eta\|$ for each $\eta \in R_2$ of the form $\eta = t(s\alpha + (1-s)\beta)$ with $t > 0$ and $s \in [0, 1]$ and $\|\eta\| = R$. Generalize this to higher dimensions with an inequality.

8 Unitary Spaces

A *unitary space* is a vector space over the field of complex numbers for which a unified presentation of real and complex an inner product is given. See [1] for inner product spaces.

Let E_n be an n -dimensional unitary space. The *Gram matrix* of the set $\{\xi_1, \dots, \xi_k\} \subset E_n$ is defined by

$$G = \begin{pmatrix} (\xi_1, \xi_1) & \cdots & (\xi_1, \xi_k) \\ \vdots & & \vdots \\ (\xi_k, \xi_1) & \cdots & (\xi_k, \xi_k) \end{pmatrix}$$

One can show that G is nonnegative, in general, and that G is and only if $\{\xi_1, \dots, \xi_k\}$ is a linearly independent set. Hence the determinant of G , denoted by $g(\xi_1, \dots, \xi_k)$, is a nonnegative real number.

One can now define perpendicular equisectors, altitudes, centroids, medians, and angle equisectors as was done for the real inner product spaces R_n and attempt to carry out a program as in Sections 3 - 7. See Exercise 8.1.

We now consider the vector space of linear transformations on E_n , denoted by $L(E_n)$. If A is a linear transformation on E_n , then the *adjoint* of A is the linear transformation A^* on E_n , defined by

$$(A\xi, \eta) = (\xi, A^*\eta) \text{ for every } \xi, \eta \in E_n$$

For two linear transformations, A and B , on E_n , we define the **-product* of A and B to be the linear transformation $A * B$ on E_n . The *-product can be viewed as a generalization of the usual inner product on the set of linear transformations on E_1 , i.e., on the complex numbers. If $n > 1$, the *-product is

not an inner product on $L(E_n)$ – it is not complex-valued! – but it does satisfy the linearity properties of an inner product, and it is true that $A * A = 0$ if and only if $A = 0$. Two linear transformations A and B on E_n are **-orthogonal* if $A * B = 0$ (and hence $B * A = 0$).

We define the *Gram determinant* of $A, B \in L(E_n)$ by

8.1 Equation

$$g(A, B) = \max(\|\xi\| = 1) \det \begin{pmatrix} (A * A\xi, \xi) & (B * A\xi, \xi) \\ (A * B\xi, \xi) & (B * B\xi, \xi) \end{pmatrix}$$

It follows from the definition of the adjoint that

$$g(A, B) = \max(\|\xi\| = 1) g(A\xi, B\xi)$$

Note that

8.2 Equation

$$g(A\xi, B\xi) = \|A\xi\|^2 \|B\xi\|^2 - |(A\xi, B\xi)|^2$$

is a nonnegative real number; that $\{\xi \in E_n \mid \|\xi\| = 1\}$ is a compact set; and thus that the maximum exists, and so $g(A, B)$ is well-defined by (8.1).

Note from (8.2) that

$$G(A\xi, B\xi) \leq \|A\xi\|^2 \|B\xi\|^2 \text{ for all } \xi \in E_n$$

and thus that

$$g(A, B) \leq \max(\|\xi\| = 1) \|A\xi\|^2 \|B\xi\|^2 \leq \|A\|_H^2 \|B\|_H^2$$

where $\|C\|_H = \max(\|\xi\| = 1) \|C\xi\|$ is the *Hilbert norm* of $C \in L(E_n)$. As a consequence of the theorem of Fischer (see [1]) applied to $A * A$, we see that $\|A\|_H^2$ is equal to the largest eigenvalue of $A * A$, and hence that

8.3 Equation

$$g(A, B) \leq ab$$

where a and b are the largest eigenvalues of $A * A$ and $B * B$, respectively.

More generally, the Gram determinant of $\{A_1, \dots, A_k\} \subset L(E_n)$ is defined by

$$g(A_1, \dots, A_k) = \max(\|\xi\| = 1) g(A_1\xi, \dots, A_k\xi)$$

This is well-defined since $g(A_1\xi, \dots, A_k\xi)$ is a nonnegative real number and the maximum is attained on the compact set $\{\xi \in E_n \mid \|\xi\| = 1\}$. It is a well-known fact that the determinant of a nonnegative matrix is less than or equal to the product of the diagonal elements and hence, by the theorem of Fisher applied as before, we obtain

8.4 Equation

$$g(A_1, \dots, A_k) \leq \prod_{i=1}^k a_i$$

where a_i is the largest eigenvalue of $A_i^2 A_i$.

Exercises

1. Define perpendicular equisectors, altitudes, centroids, medians, and angle equisectors in E_n . Examine the theorems and exercises of Sections 3 - 7 in this setting. (Caution: $\beta, \alpha = \overline{(\alpha, \beta)}$.)
2. Show that $g(A_1, \dots, A_k) \neq 0$ if and only if $\{A_1, \dots, A_k\}$ is a linearly independent set in $L(E_n)$.
3. Show that equality holds in (8.3) if A and B are $*$ -orthogonal, and the intersection of the a eigenspace of A and the b eigenspace of B is nontrivial. Find necessary and sufficient conditions for equality to hold in (8.4).
4. Solve the problems on $*$ -products given in Chapter 4 of [1].
5. Define the k -dimensional Hilbert volume measure of the parallelepiped defined by $\{A_1, \dots, A_k\}$ in $L(E_n)$ to be

$$V = [g(A_1, \dots, A_k)]^{\frac{1}{2}}$$

Investigate Theorem 2.2 and the exercises of Section 2 in this setting.

6. Define perpendicular equisectors, altitudes, centroids, medians, and angle equisectors in $L(E_n)$ using the $*$ -product and the Hilbert norm. Examine the theorems and exercises of Sections 3 - 7 in this setting.

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